

George Dowling

Senior Data Engineer – Azure Databricks

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Summary

Senior Data Engineer with 10+ years of progressive experience building and modernizing enterprise data platforms, growing from foundational ETL development into leading large-scale Azure Databricks Lakehouse architectures for global organizations. I specialize in Azure-native data engineering, Delta Lake optimization, governed medallion architectures, and secure multi-environment CI/CD deployments. My work consistently improves performance, reliability, compliance posture, and cost efficiency across analytics and AI-ready data ecosystems.

I am known for disciplined engineering habits, strong documentation standards, and proactive stakeholder communication. Outside of work, I enjoy long-distance running and reading about distributed systems architecture, which continuously sharpens my endurance mindset and systems thinking approach to engineering challenges.

Education

Princeton University

Master's degree in Data Science

Jun 2015 – Apr 2017

Princeton University

Bachelor's degree in Computer Science

Sep 2011 – May 2015

Certification

- Microsoft Certified: Azure Data Engineer Associate
- Databricks Certified Data Engineer Professional
- Microsoft Certified: Azure Solutions Architect Expert
- Microsoft Certified: Azure Security Engineer Associate

Skills

- **Azure & Databricks Ecosystem:** Azure Databricks (Enterprise Workspaces), Delta Lake, Delta Live Tables, Unity Catalog, Autoloader, Databricks Workflows, MLflow integration, Databricks SQL, cluster governance and optimization
- **Cloud Platforms:** Microsoft Azure (ADLS Gen2, Azure Data Factory, Azure Functions, Azure Key Vault, Managed Identities, Azure Monitor), AWS services (S3, IAM, Glue, CloudWatch, Lambda)
- **Distributed Data Processing:** Apache Spark, PySpark, Spark SQL, Structured Streaming, Kafka, Azure Event Hub
- **Data Architecture & Modeling:** Enterprise Lakehouse Architecture, Bronze/Silver/Gold Medallion Design, Dimensional Modeling, 3NF, Data Vault concepts, Logical and Physical Data Modeling, Semantic Layer Enablement
- **CI/CD & DevOps:** Azure DevOps, GitHub Actions, JFrog Artifactory, Infrastructure-as-Code principles, environment promotion strategies, automated testing frameworks, release governance
- **Governance & Security:** Unity Catalog RBAC, fine-grained access control, data masking, lineage tracking, audit logging, encryption strategies, compliance-ready data architecture, disaster recovery planning
- **Data Quality & Observability:** Metadata-driven ingestion frameworks, CDC-based processing, automated validation frameworks, anomaly detection patterns, monitoring and alerting, SLA/SLO tracking
- **Soft Skills:** Executive communication, cross-functional leadership, technical mentorship, stakeholder alignment, documentation discipline, strategic roadmap contribution, Agile collaboration, risk management, change management

Work Experience

Senior Data Engineer

Hulu | Santa Monica, CA

Jul 2023 – Present

- Lead the enterprise architecture and implementation of an Azure Databricks Delta Lake Lakehouse platform using Bronze/Silver/Gold medallion architecture to deliver governed, analytics-ready, and AI-enabled datasets across content intelligence and security domains.
- Designed and implemented highly scalable ingestion and transformation pipelines supporting both batch and structured streaming workloads via Kafka and Azure Event Hub, enabling near-real-time fraud detection and operational monitoring use cases.
- Engineered advanced Delta Lake optimization strategies leveraging ACID guarantees, schema enforcement, partition pruning, Z-ordering, and time travel capabilities, significantly reducing reprocessing overhead and improving recovery reliability across production pipelines.
- Built metadata-driven ingestion frameworks using Autoloader and CDC-based incremental processing, reducing new data source onboarding timelines from multiple weeks to several days while maintaining strict governance controls.
- Developed complex PySpark and Spark SQL transformation pipelines with advanced query tuning, adaptive execution strategies, and workload isolation, improving end-to-end processing performance by over 28% across high-volume datasets.
- Architected and managed enterprise Databricks workspaces, clusters, job orchestration patterns, and compute optimization strategies, implementing right-sizing and auto-scaling policies that reduced monthly infrastructure spend without compromising SLO adherence.
- Implemented centralized governance using Unity Catalog with fine-grained RBAC policies, secure data sharing models, data classification enforcement, and lineage tracking to strengthen auditability and compliance posture across regulated datasets.
- Designed secure Azure-native architectures leveraging Managed Identities, Azure Key Vault integration, private networking, and role-based access models to ensure secure secret management and resilient data platform operations.
- Established enterprise-grade data quality frameworks incorporating automated validation rules, anomaly detection models, defect tracking integration with JIRA, and proactive monitoring to reduce escaped data issues and increase stakeholder trust.
- Built CI/CD pipelines for Databricks notebooks, jobs, and infrastructure components using Azure DevOps and GitHub, automating testing, environment promotion, and rollback procedures across development, staging, and production environments.
- Contributed to AI/ML data enablement by preparing feature-ready curated datasets and optimizing storage layouts for downstream model training, experimentation workflows, and LLM-adjacent analytics initiatives.
- Mentor junior engineers through structured code reviews, architectural design discussions, and documentation standards while actively translating complex technical trade-offs into clear executive-level communication.

Senior Data Engineer

IBM | Armonk, NY

Jan 2019 – Jun 2023

- Led enterprise lakehouse modernization programs migrating legacy on-premises data warehouses and monolithic ETL frameworks into scalable Azure Databricks and Delta Lake architectures aligned with governance-first design principles.
- Designed and implemented scalable ingestion, transformation, and curation frameworks leveraging Azure Data Factory, ADLS Gen2, and Spark-based processing to support both batch analytics and near-real-time operational reporting.
- Built multi-tenant data processing frameworks supporting segregated environments across development, staging, and production while maintaining consistent deployment, monitoring, and governance standards.

- Developed highly optimized PySpark transformation pipelines applying advanced partition strategies, broadcast joins, query plan analysis, and adaptive execution, improving latency and throughput across enterprise datasets.
- Implemented Delta Lake schema evolution and enforcement controls to improve traceability, reduce defect investigation time, and maintain strong data integrity guarantees across regulated compliance workloads.
- Designed and automated CI/CD pipelines for Databricks artifacts using Azure DevOps, GitHub, and JFrog Artifactory, reducing release cycles and improving deployment consistency across global client environments.
- Established centralized monitoring and observability frameworks integrating job metrics, failure detection alerts, and SLA dashboards to proactively manage operational risks and ensure high-availability platform performance.
- Implemented Unity Catalog governance standards including fine-grained RBAC, lineage tracking, and secure data sharing, enabling enterprise-wide auditability and controlled cross-domain data access.
- Partnered closely with QA, security, and compliance teams to validate transformation accuracy, enforce encryption standards, and ensure alignment with regulatory frameworks and enterprise risk policies.
- Contributed to architectural roadmaps, capacity planning, and cost optimization strategies for Azure-based data platforms supporting large-scale analytics and reporting programs.
- Facilitated executive-level stakeholder workshops to align technical modernization initiatives with strategic business objectives and long-term data platform vision.

Data Engineer

Symufolk | NYC, NY

May 2018 – Dec 2018

- Developed production-grade ETL pipelines using Python and SQL, integrating operational and transactional datasets into structured analytics layers supporting enterprise reporting and decision-making processes.
- Designed dimensional and normalized data models to support scalable reporting architectures while ensuring semantic clarity and consistent KPI definitions across stakeholder groups.
- Implemented incremental and CDC-style ingestion frameworks reducing reliance on full-refresh patterns and improving reliability and operational efficiency across daily batch workloads.
- Applied advanced query optimization and indexing strategies to improve data retrieval performance and enhance dashboard responsiveness during peak reporting windows.
- Introduced foundational monitoring and alerting mechanisms to detect pipeline failures early, reducing downtime and improving overall data delivery reliability.
- Contributed to early cloud adoption initiatives leveraging AWS S3 for object storage, IAM for secure access control, and CloudWatch for job monitoring within hybrid deployment models.
- Assisted in transitioning select workloads into AWS Glue and Lambda-based orchestration experiments, gaining foundational exposure to serverless data processing architectures.
- Built automated validation scripts to compare source-to-target transformations, increasing trust in curated datasets and reducing manual reconciliation efforts.
- Collaborated with cross-functional product and analytics teams to translate evolving business requirements into scalable, maintainable data engineering solutions.
- Documented data lineage, transformation logic, and architectural decisions to improve maintainability and knowledge transfer within a growing engineering team.

Data Engineer

SwagUp | Piscataway, NJ

May 2017 – Apr 2018

- Designed and implemented ETL pipelines consolidating finance, operations, and transactional datasets into curated analytics-ready structures supporting executive reporting and KPI tracking.
- Developed reusable ingestion frameworks incorporating incremental processing and data validation checkpoints to improve daily refresh efficiency and reduce compute waste.

- Modeled core business entities using dimensional techniques to enable consistent reporting semantics and support scalable BI consumption patterns.
- Implemented structured data quality validation rules to identify anomalies in financial and operational datasets before downstream publication.
- Optimized SQL-based transformations and storage layouts to improve reporting performance and ensure reliable dashboard refresh cycles under heavy business usage.
- Collaborated closely with finance and operations stakeholders to clarify requirements, prioritize feature delivery, and communicate architectural trade-offs in Agile development environments.
- Introduced documentation standards and lightweight version control practices to improve code maintainability and knowledge continuity across the engineering team.
- Supported platform stability through systematic troubleshooting, job dependency analysis, and performance monitoring of scheduled workloads.
- Assisted in early cloud integration efforts to improve scalability of analytics storage and processing environments.
- Contributed to iterative modernization of legacy scripts into more modular, maintainable data processing components aligned with emerging best practices.